

Eigenvalue Estimation via the Rodeo Algorithm: Application to Ising and Discrete Dirac Hamiltonians

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Abstract

The Rodeo algorithm is a stochastic quantum-circuit filter that estimates Hamiltonian eigenvalues with exponential suppression of off-resonance components. We implement it in Qiskit and benchmark it on the transverse-field Ising model ($L=4$) across three progressively realistic noise regimes: noiseless simulation, custom depolarising channels, and the calibrated IBM Manila hardware profile. A systematic depth-reduction study (Trotter steps $n=20 \rightarrow 3$) combined with a t_{rms} resolution analysis identifies the parameter window compatible with real hardware execution. The resulting circuit is executed on the `ibm_kingston` processor via the IBM Quantum Platform [4], with and without dynamical-decoupling and Pauli-twirling error mitigation. The same Rodeo circuit—without structural modification—is then applied to the discrete Dirac, Heisenberg XXX, and 2D lattice Hamiltonians, and a parameter-sweep mode detects the ferromagnetic-to-paramagnetic transition in the TFIM. Source code: https://github.com/ArthurFCSMello/Rodeo_project.

1 Introduction

Extracting eigenvalues of many-body Hamiltonians is among the most computationally demanding tasks in physics and chemistry, granting access to ground-state energies, excitation spectra, and phase structure [6, 7]. Quantum Phase Estimation (QPE) solves this problem in principle, but its $O(1/\varepsilon)$ circuit depth renders it impractical on current noisy intermediate-scale quantum (NISQ) hardware.

The Rodeo algorithm [1] offers an alternative: a stochastic spectral filter whose success probability is exponentially peaked at true eigenvalues after N ancilla measurement cycles, at a cost of $O[(\log \varepsilon)^2/(p\varepsilon)]$ for error ε and initial-state overlap $p = |\langle \psi_0 | E_n \rangle|^2$. Related quantum eigenvalue approaches have been studied in [5].

This report presents a complete implementation-to-hardware pipeline, developed within the quantum-computing curriculum at CentraleSupélec. Three contributions are made, in decreasing order of specificity:

1. **Noise–depth characterisation.** We quantify how depolarising noise, the IBM Manila hardware profile [4], circuit depth (N cycles, n Trotter steps), and the filter bandwidth t_{rms} jointly determine whether the Rodeo signal survives. The analysis identifies the parameter regime ($n=3$, $N=5$, $t_{\text{rms}}=1$) that makes real-hardware execution feasible.
2. **Hardware execution.** The optimised circuit is run on the IBM Quantum Platform [4], with and without error mitigation (dynamical decoupling, Pauli twirling), providing a direct comparison between simulated and measured spectra.
3. **Hamiltonian generality.** The same Rodeo circuit is applied—unchanged—to the transverse-field Ising model (1D and 2D), the Heisenberg XXX chain, the discrete Dirac Hamiltonian (1D and 2D, including time-domain matter–antimatter propagation), and the 2D harmonic oscillator, validating the algorithm’s model-independence.

The central theme throughout is the *depth–accuracy trade-off*: more Rodeo cycles and finer Trotterisation improve the ideal spectral resolution, but every additional gate exponentially degrades the measured signal in the presence of noise. Navigating this trade-off is the principal engineering challenge addressed here.

2 Theoretical Framework

2.1 The Rodeo Filter

Consider a Hamiltonian H with eigenstates $|E_n\rangle$ and eigenvalues E_n . An ancilla qubit, prepared in superposition and coupled to the system via controlled time evolution $CU(t) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{-iHt}$, accumulates a

target-energy phase $e^{iE_{\text{target}}t}$ before a final Hadamard and measurement. The probability of measuring the ancilla in $|1\rangle$ is

$$P_k(E_n) = \cos^2\left[\frac{(E_n - E_{\text{target}})t_k}{2}\right]. \quad (1)$$

After N independent cycles with random times $\{t_k\}$ drawn from $\mathcal{N}(0, t_{\text{rms}})$, the joint success probability is

$$P(E_n) = \prod_{k=1}^N \cos^2\left[\frac{(E_n - E_{\text{target}})t_k}{2}\right]. \quad (2)$$

This product peaks sharply at $E_{\text{target}} = E_n$ and is exponentially suppressed elsewhere, yielding a stochastic spectral filter whose resolution improves with N .

2.2 n -Dimensional Generalisation

For a separable Hamiltonian $H = \sum_{j=1}^d H_j$ with $[H_j, H_\ell] = 0$ and common product eigenbasis $|\mathbf{n}\rangle = |n_1\rangle \otimes \dots \otimes |n_d\rangle$, $H_j |n_j\rangle = E_{n_j}^{(j)} |n_j\rangle$, the single-cycle filter factor for eigenmode $\mathbf{n}$ reads

$$F_k(\mathbf{n}; \mathbf{E}_{\text{tgt}}) = \prod_{j=1}^d \cos^2\left[\frac{(E_{n_j}^{(j)} - E_{\text{tgt},j})t_j^{(k)}}{2}\right]. \quad (3)$$

After N cycles, with overlap weights $w_{\mathbf{n}} = |\langle \mathbf{n} | \psi_0 \rangle|^2$,

$$P_N(\mathbf{E}_{\text{tgt}}) = \sum_{\mathbf{n}} w_{\mathbf{n}} \prod_{k=1}^N F_k(\mathbf{n}; \mathbf{E}_{\text{tgt}}). \quad (4)$$

If ψ_0 factorises, $w_{\mathbf{n}} = \prod_j w_{n_j}^{(j)}$, the filter itself factorises:

$$P_N(\mathbf{E}_{\text{tgt}}) = \prod_{j=1}^d \left[\sum_{n_j} w_{n_j}^{(j)} \prod_{k=1}^N \cos^2\left(\frac{(E_{n_j}^{(j)} - E_{\text{tgt},j})t_j^{(k)}}{2}\right) \right]. \quad (5)$$

The cost therefore scales linearly in d rather than exponentially in Hilbert-space dimension—the regime exploited in our 2D harmonic-oscillator and 2D Dirac benchmarks.

2.3 Pauli Operators

We adopt the standard Pauli matrices [6]:

$$\sigma_x = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

The shorthand X_i denotes $I^{\otimes(i-1)} \otimes \sigma_x \otimes I^{\otimes(L-i)}$; $Z_i Z_{i+1}$ is the tensor product $Z_i \otimes Z_{i+1}$, measuring the parity correlation between neighbouring qubits.

2.4 Transverse-Field Ising Model

The Hamiltonian on L sites with open boundary conditions is

$$H_{\text{Ising}} = -J \underbrace{\sum_{i=1}^{L-1} Z_i Z_{i+1}}_{H_{\text{ZZ}}} - h \underbrace{\sum_{i=1}^L X_i}_{H_X}, \quad (7)$$

with coupling $J = 1.0$ and transverse field $h = 1.5$ throughout.

Worked example: $L=2$. For two qubits the Hamiltonian reduces to $H_{L=2} = -J Z_1 Z_2 - h(X_1 + X_2)$, i.e. the 4×4 matrix

$$H_{L=2} = \begin{pmatrix} -J & -h & -h & 0 \\ -h & J & 0 & -h \\ -h & 0 & J & -h \\ 0 & -h & -h & -J \end{pmatrix}. \quad (8)$$

At $J=1, h=1.5$ the eigenvalues are $\{-2.80, -1.00, +0.80, +3.00\}$. The Rodeo filter recovers these four values by scanning E_{tgt} and detecting peaks in $P(E_{\text{tgt}})$, without classical diagonalisation.

Since $[H_{\text{ZZ}}, H_X] \neq 0$, the unitary e^{-iHt} cannot be factored exactly into native gates.

Trotterisation. The first-order Lie–Trotter formula [3] splits H into its non-commuting parts at time step $\delta t = t/n$:

$$e^{-iHt} = e^{-i(H_{\text{ZZ}} + H_X)t} \approx \left(e^{-iH_X \delta t} e^{-iH_{\text{ZZ}} \delta t} \right)^n. \quad (9)$$

The accumulated error is $O(t^2/n)$. Each factor decomposes into native gates:

- $e^{-iH_X \delta t} = \prod_i e^{ih \delta t X_i}$: single-qubit $R_x(2h \delta t)$ rotations, controlled by the ancilla via CR_x gates.
- $e^{-iH_{\text{ZZ}} \delta t} = \prod_i e^{iJ \delta t Z_i Z_{i+1}}$: the standard CNOT sandwich $\text{CNOT}_{i,i+1} CR_z(2J \delta t) \text{CNOT}_{i,i+1}$.

2.5 1D Discrete Dirac Hamiltonian

The continuum 1D Dirac equation [7] for mass m reads $i\partial_t \psi = (c\alpha \hat{p} + \beta mc^2)\psi$, with $\{\alpha, \beta\} = 0$, $\alpha^2 = \beta^2 = I$. We set $\alpha = \sigma_z$, $\beta = \sigma_x$, and $c = \hbar = 1$.

Lattice discretisation. On L sites with spacing a , the momentum operator becomes $\hat{p} \rightarrow (\hat{S} - \hat{S}^\dagger)/(2ia)$, where $\hat{S}$ is the cyclic shift:

$$\hat{S} |n\rangle = |(n+1) \bmod L\rangle, \quad \hat{S}^\dagger |n\rangle = |(n-1) \bmod L\rangle. \quad (10)$$

Explicitly,

$$\hat{S} = \sum_{n=0}^{L-1} |(n+1) \bmod L\rangle \langle n| = \begin{pmatrix} 0 & 0 & \dots & 1 \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 1 & 0 \end{pmatrix}_{L \times L}. \quad (11)$$

The discrete Dirac Hamiltonian on $\mathcal{H} = \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{spin}}$ (dimension $2L$) is

$$H_D = \underbrace{\frac{\sigma_z}{2ia} \otimes (\hat{S} - \hat{S}^\dagger)}_{\text{kinetic hopping}} + \underbrace{m \sigma_x \otimes I_L}_{\text{mass term}}. \quad (12)$$

The mass term couples the two spinor components on-site; the kinetic term hops each component to its neighbours with amplitude $\pm 1/(2a)$ and a σ_z -governed spinor flip.

Qubit encoding. Position $n \in \{0, \dots, L-1\}$ is stored in $\lceil \log_2 L \rceil$ qubits (binary encoding); one additional qubit carries the spinor index, for a total of $\lceil \log_2 L \rceil + 1$ qubits. The shift operator is implemented as a controlled binary increment via multi-controlled X (MCX) gates (Listing 3).

Dispersion relation. With periodic boundaries and discrete momenta $k_j = 2\pi j/(La)$, the exact eigenvalues are

$$E_{\pm}(k_j) = \pm \sqrt{\left(\frac{\sin(k_j a)}{a}\right)^2 + m^2}, \quad (13)$$

reproducing the continuum relativistic dispersion $E = \pm \sqrt{p^2 + m^2}$ in the long-wavelength limit. The $+$ ($-$) branch corresponds to matter (antimatter) states.

2.6 2D Discrete Dirac Hamiltonian

On a 2D lattice ($L_x \times L_y$ sites) two anticommuting kinetic operators are needed, representable with three Pauli matrices [7]:

$$\begin{aligned} H_D^{2D} = & \frac{\sigma_x}{2ia} \otimes (\hat{S}_x - \hat{S}_x^\dagger) \otimes I_y \\ & + \frac{\sigma_y}{2ia} \otimes I_x \otimes (\hat{S}_y - \hat{S}_y^\dagger) \\ & + m \sigma_z \otimes I_x \otimes I_y, \end{aligned} \quad (14)$$

where σ_x (σ_y) couples spinor components during x -hops (y -hops) and σ_z provides the mass gap. The 2D eigenvalues are

$$E_{\pm}(\mathbf{k}) = \pm \sqrt{\frac{\sin^2(k_x a)}{a^2} + \frac{\sin^2(k_y a)}{a^2} + m^2}. \quad (15)$$

3 Implementation

All circuits are built with Qiskit v2 [2] and executed on the **AerSimulator** backend. Figure 1 shows the single-ancilla Rodeo cycle.

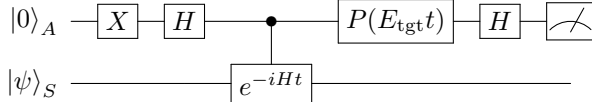


Figure 1: Illustrative quantum circuit for a single Rodeo cycle. The ancilla (top) controls the time evolution of the system register (bottom) over a sampled duration t , before accumulating the target energy phase and being measured. For multi-cycle operation on a single ancilla, a reset is applied returning the qubit to $|0\rangle$, followed by an X gate to restore it to $|1\rangle$ before the next cycle.

The controlled Trotterised evolution for the Ising model is given in Listing 1.

Single-ancilla reformulation. The original Rodeo protocol allocates one ancilla per cycle, so an N -cycle run requires N ancillae measured simultaneously. On a classical state-vector simulator this rapidly becomes intractable: the enlarged register scales as 2^{L+N} , pushing simulation times to hours for $L=4$, $N \geq 5$. We therefore adopt a single-ancilla design with mid-circuit measurement and unconditional reset (Listing 2): after each cycle the ancilla is measured, reset to $|0\rangle$, and flipped to $|1\rangle$ via an X gate, keeping the qubit count fixed regardless of N .

Circuit depth. The two-qubit gate count scales as $O(N \cdot n \cdot L)$. For $L=4$, $n=20$, $N=10$, each circuit exceeds 1 200 CNOTs, which explains the severe signal degradation under realistic noise. The Dirac circuit is even deeper: the shift operator requires MCX gates (Listing 3), each decomposing into dozens of CNOTs on restricted topologies.

Simulation parameters. Noiseless studies (Sec. 4.1) use up to 10 000 shots per scan point over $E \in [-8, 8]$. Noisy runs (Secs. 4.3–4.8) adapt shot count and scan density to each regime. Random times are drawn from $\mathcal{N}(0, t_{\text{rms}})$ with $t_{\text{rms}} \in \{0.75, \dots, 5\}$.

Noise models. Three progressively realistic scenarios are studied:

1. **Noiseless:** perfect gates; only shot noise remains.
2. **Depolarising:** uniform channel ($p_{1q}=0.001$, $p_{2q}=0.01$) after every gate, isolating incoherent noise.
3. **IBM Manila profile:** calibrated T_1/T_2 relaxation, readout asymmetries, and gate-dependent errors from the **FakeManilaV2** backend [2, 4].

The Manila simulation uses the single-ancilla circuit (Listing 2) to fit the five-qubit topology. In Sec. 4.8 the same circuit runs on real IBM hardware.

4 Results

All Ising-model results use $L=4$, $J=1.0$, $h=1.5$, with the energy axis scanned over $E \in [-8, 8]$. Black stems mark exact eigenvalues from classical diagonalisation of the 16×16 matrix.

4.1 Noiseless Ising Spectrum with Trotterisation ($n=20$)

The unitary e^{-iHt} is decomposed via Eq. (9) with $n=20$ steps and $t_{\text{rms}}=5$, yielding negligible Trotter error. Figure 2 illustrates the filter sharpening as N increases: at low N the response is broad; by $N=30$ the peaks align with the true eigenvalues to within the scan resolution.

Figures 3 and 4 compare $N=5$ and $N=10$ at the same shot count (5 000), isolating the effect of the cycle number. At $N=5$ the dominant eigenvalues near $E \approx -5.0$, -3.4 , and -2.1 are resolved; doubling to $N=10$ suppresses the inter-peak floor and reveals weaker features at positive energies.

4.2 Influence of Shot Count on Spectral Quality

Each point of $P(E_{\text{tgt}})$ is estimated from a finite number of circuit executions; the associated statistical uncertainty scales as $1/\sqrt{N_{\text{shots}}}$. Figure 5 varies the shot count at fixed $N=5$. At 100 shots the dominant peaks are discernible but noisy; at 10 000 shots the curve is smooth and the main peaks approach the exact positions. Spurious features persist near $E \approx 5$ and $E \approx 7.5$, indicating that $N=5$ cycles do not yet provide sufficient destructive interference to suppress all non-eigenvalue energies.

```

1 # Transverse-field term:  $-h X_i$ 
2 for i in range(num_qubits):
3     qc.crx(-2 * h * dt, anc, sys[i])
4 # Ising coupling term:  $-J Z_i Z_{i+1}$ 
5 for i in range(num_qubits - 1):
6     qc.cx(sys[i], sys[i+1])
7     qc.crz(-2 * J * dt, anc, sys[i+1])
8     qc.cx(sys[i], sys[i+1])

```

Listing 1: Controlled Trotter step for the Ising Hamiltonian (one slice).

```

1 # Start ancilla in  $|1\rangle$ 
2 qc.x(qr_anc)
3
4 for m in range(cycles):
5     t, dt = t_list[m], t_list[m] / trotter_steps
6     qc.h(qr_anc)
7
8     # ... Trotter steps inserted here ...
9
10    qc.p(E_target * t, qr_anc)
11    qc.h(qr_anc)
12    qc.measure(qr_anc, cr_anc[m])
13
14    # Always reset: return to  $|0\rangle$  for any output and then apply x-gate
15    qc.reset(qr_anc)
16    qc.x(qr_anc) # Turns the state to  $|1\rangle$  always

```

Listing 2: Single-ancilla Rodeo filter with mid-circuit measurement and reset for hardware execution.

```

1 # Controlled forward hopping
2 qc.crz(-2 * hop, anc, spin)
3 # Multi-controlled position increment
4 ctrl_list = [anc] + list(pos[:-1])
5 qc.mcx(ctrl_list, pos[-1])
6 for i in range(len(pos) - 2, 0, -1):
7     ctrl = [anc] + list(pos[:i])
8     qc.mcx(ctrl, pos[i])
9 qc.cx(anc, pos[0])

```

Listing 3: Increment logic for the Dirac kinetic hopping term.

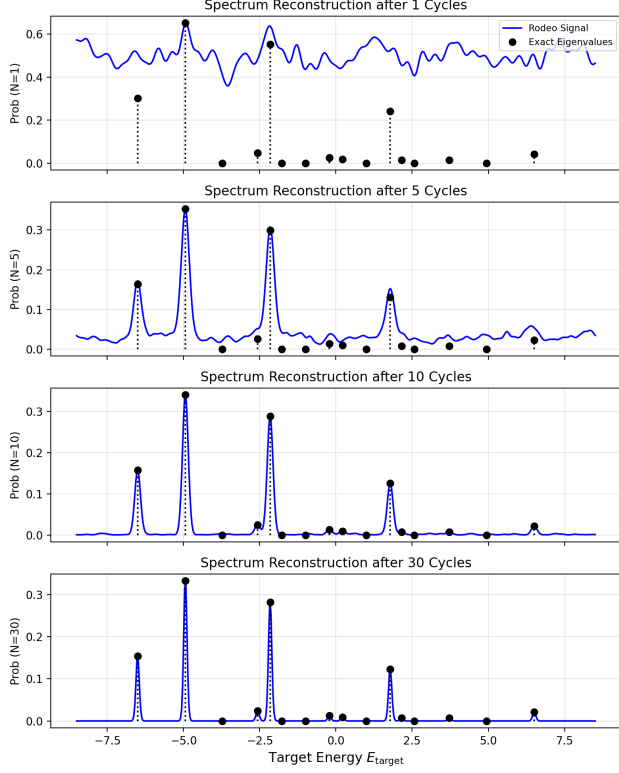


Figure 2: Evolution of $P(E_{\text{tgt}})$ with increasing cycle count N for the initial state $|0000\rangle$. At low N the curve is broad; as N grows the peaks converge exponentially onto the exact eigenvalues (dashed stems).

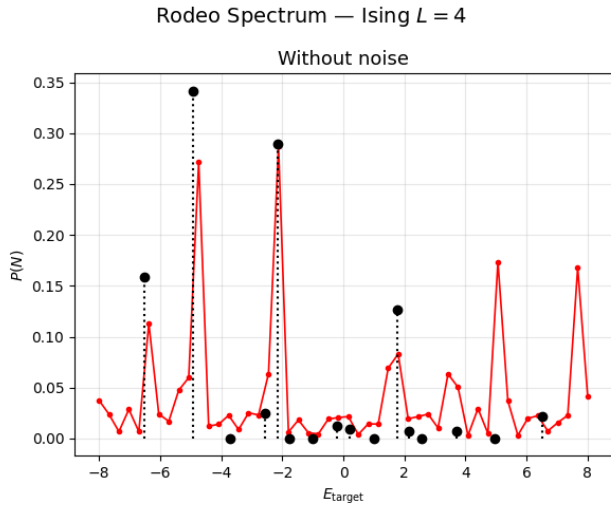


Figure 3: Noiseless Rodeo spectrum, $N=5$ cycles, 5000 shots, $n=20$ Trotter steps, $t_{\text{rms}}=5$, 50 scan points.

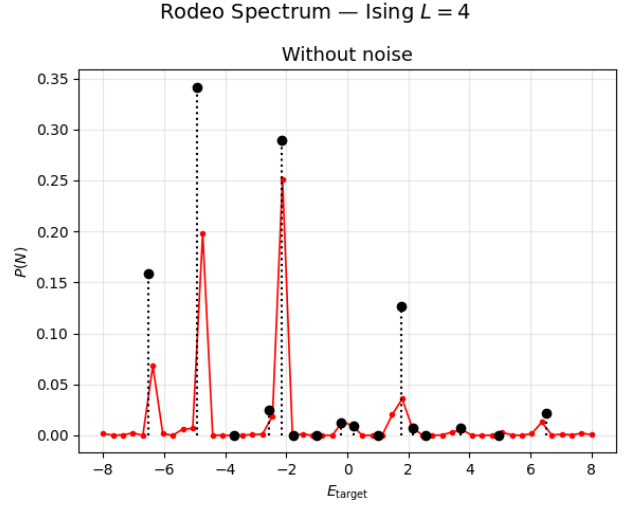


Figure 4: Noiseless Rodeo spectrum, $N=10$ cycles, 5000 shots. The additional cycles sharpen peaks and reveal subdominant eigenvalues that were buried in the $N=5$ background.

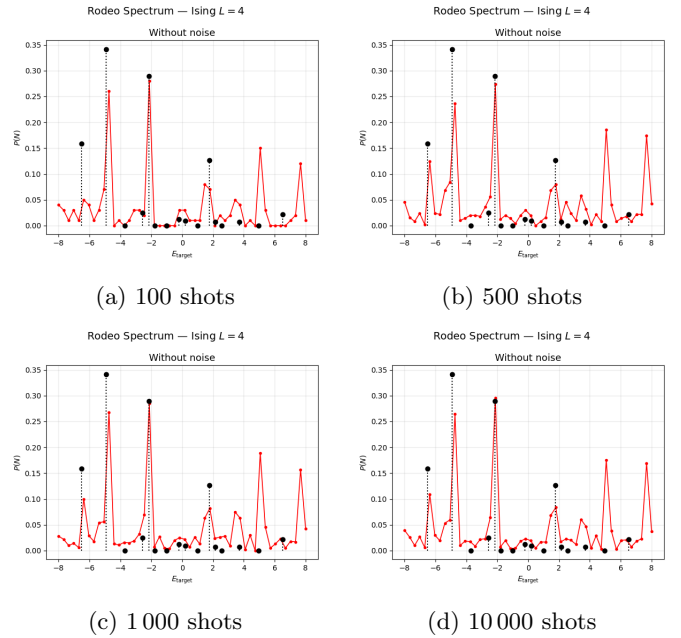


Figure 5: Shot-count convergence (noiseless, $N=5$ cycles, $t_{\text{rms}}=5$, 50 scan points). Increasing N_{shots} reduces statistical noise and reveals progressively finer spectral structure. Black dots: exact eigenvalues.

4.3 Depolarising Noise

After each gate the single-qubit depolarising channel replaces the state ρ with the maximally mixed state with probability p :

$$\mathcal{E}_{\text{depol}}(\rho) = (1-p)\rho + p \frac{I}{2}. \quad (16)$$

For two-qubit gates ρ mixes with $I/4$. The net effect on the Rodeo filter is a progressive flattening of $P(E_{\text{tgt}})$ toward $1/2^N$, because every noisy gate degrades the ancilla-system coherence on which the filter relies.

Varying error probability at fixed cycles. Figure 6 scans p from 10^{-5} to the default level ($p_{1q}=0.001$, $p_{2q}=0.01$) at $N=10$. At $p=10^{-5}$ the spectrum is indistinguishable from the noiseless case; at $p=10^{-3}$ only the ground-state peak survives; at the default level the signal is flat. The coherent survival probability $(1-p)^G$ with $G \sim 2000$ two-qubit gates explains this exponential sensitivity.

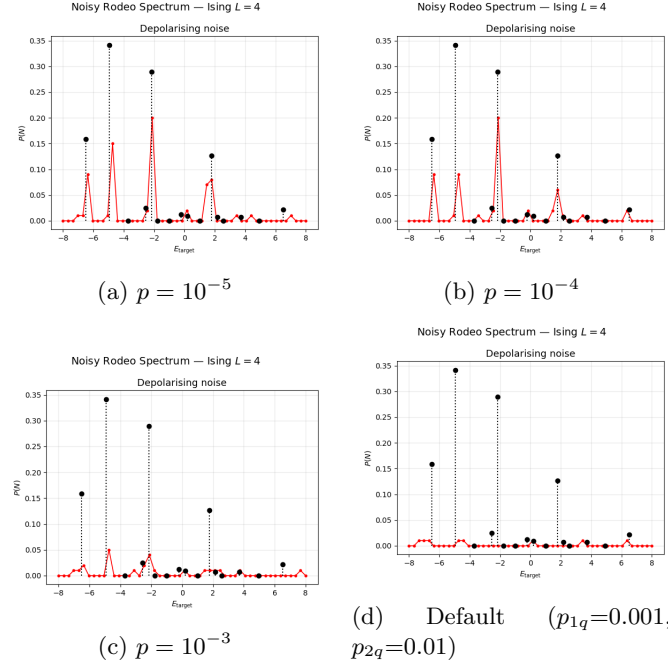


Figure 6: Depolarising noise: $N=10$ cycles, 100 shots, $n=20$ Trotter steps. Increasing the per-gate error probability progressively flattens the spectral signal. Black dots: exact eigenvalues.

Varying cycle count at fixed error rate. Figure 7 fixes $p=10^{-4}$ and increases $N \in \{3, 5, 10\}$. Each additional cycle appends $n \times 3L$ controlled gates, so $G \propto N \cdot n \cdot L$. At $N=3$ the filter is broad but noise-tolerant; at $N=10$ the ideal resolution is higher but accumulated decoherence erodes the signal—quantifying the *resolution vs. resilience* trade-off.

Circuit depth — the central trade-off. The two-qubit gate count is approximately

$$G_{2q} \approx N \cdot n \cdot (3L - 2), \quad (17)$$

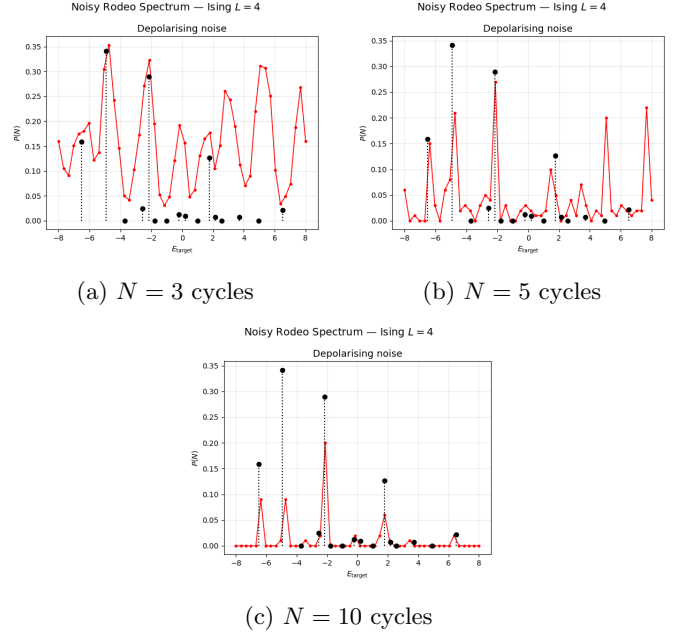


Figure 7: Depolarising noise at fixed $p=10^{-4}$: effect of increasing the number of Rodeo cycles. More cycles sharpen the ideal filter but linearly increase circuit depth, amplifying the accumulated noise.

yielding $G_{2q} \approx 2000$ for $N=10$, $n=20$, $L=4$. The error-free survival probability is $(1-p_{2q})^{G_{2q}}$; at $p_{2q}=0.01$ this gives $(0.99)^{2000} \approx 2 \times 10^{-9}$, explaining the total loss of signal. This establishes the **depth-accuracy dilemma**: increasing N or n improves the ideal filter but exponentially degrades the measured signal. All subsequent parameter studies (Secs. 4.4–4.8) navigate this trade-off.

4.4 IBM FakeManila Hardware-Profile Simulation

The FakeManilaV2 backend [2, 4] reproduces the calibrated error rates, T_1/T_2 relaxation, readout asymmetries, and coupling-map constraints of the 5-qubit IBM Falcon r1 processor. Unlike the uniform depolarising channel, this profile includes gate- and qubit-dependent errors together with thermal relaxation during idle periods.

Figure 8 shows the FakeManila spectrum at $N \in \{2, 5, 10, 15\}$ with $n=20$. The signal is flat at all cycle counts: with $T_2 \sim 100 \mu\text{s}$ and gate times $\sim 0.3 \mu\text{s}$, circuits beyond ~ 300 layers exceed the coherence window. Depth reduction is therefore mandatory.

4.5 Reducing Circuit Depth: $n=3$ Trotter Steps

Reducing n from 20 to 3 introduces Trotter error but cuts G_{2q} by a factor of ~ 7 , from ~ 2000 to ~ 300 for $N=5$. Figure 9 shows the noiseless $n=3$ spectrum at $N=4$ and $N=5$ (with $t_{\text{rms}}=5$). Some peaks correlate with exact eigenvalues, but the filter probability for true and spurious peaks is comparable; in several cases false peaks exceed the true ones. This ambiguity motivates the t_{rms} variation study

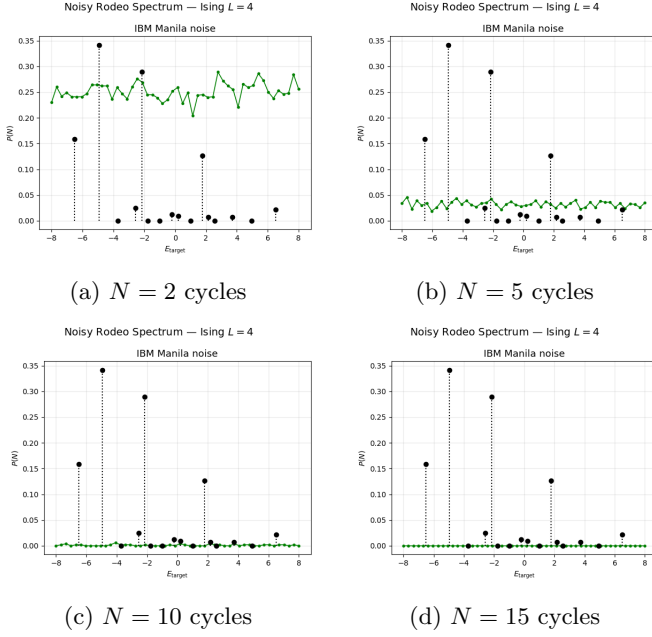


Figure 8: IBM FakeManila noise profile ($n=20$ Trotter steps). The realistic hardware errors flatten the spectrum at all cycle counts, indicating that $n=20$ exceeds the device coherence budget. Green: simulated spectrum; black dots: exact eigenvalues.

in the next section.

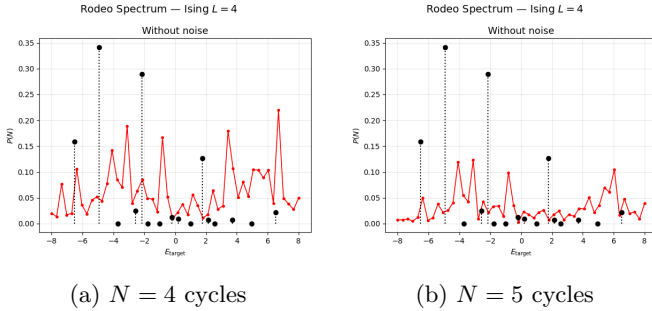


Figure 9: Noiseless Rodeo spectrum with reduced Trotter depth $n=3$, $t_{\text{rms}}=5$, 2000 shots. Some low-lying peaks remain aligned with exact eigenvalues, but the filter is not strong enough to distinguish true peaks from spurious ones at this resolution.

4.6 Energy Resolution vs. Filter Bandwidth (t_{rms})

The filter width obeys the time-energy uncertainty relation

$$\Delta E \sim \frac{1}{t_{\text{rms}}}. \quad (18)$$

Large t_{rms} sharpens peaks but lengthens controlled evolutions; small t_{rms} broadens peaks but keeps depth manageable. Figure 10 scans $t_{\text{rms}} \in \{0.75, 1.0, 1.25, 1.5\}$ at $n=3$, $N=5$. At $t_{\text{rms}}=0.75$ the peaks overlap; at $t_{\text{rms}}=1.5$ they are narrower and better separated, though the maximum

peak probability decreases by $\sim 10\%$. The value $t_{\text{rms}}=1.0$ is selected for hardware as a compromise between resolution and circuit depth.

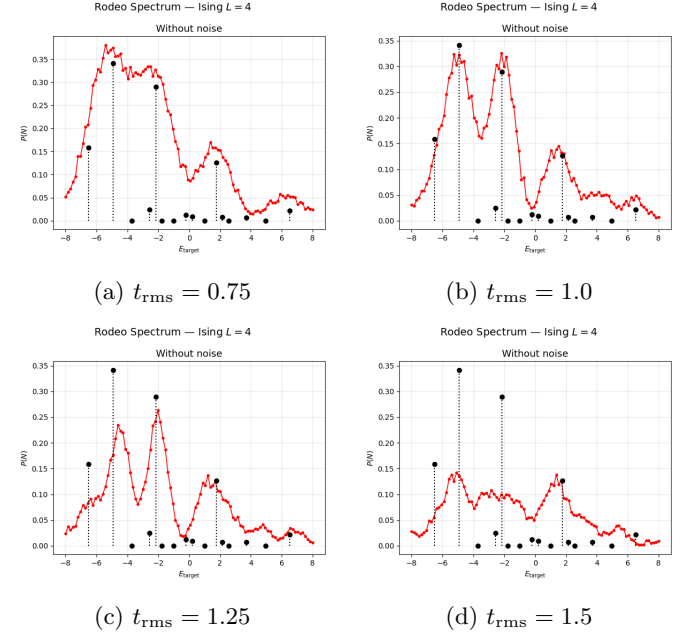


Figure 10: Effect of t_{rms} on peak width (noiseless, $n=3$, $N=5$, 2000 shots, 100 scan points). Smaller t_{rms} broadens the peaks (lower energy resolution); larger t_{rms} sharpens them but requires longer evolution times and hence deeper circuits.

4.7 IBM FakeManila Simulation with Reduced Depth

With the optimised parameters ($n=3$, $N=5$, $t_{\text{rms}}=1.0$, 2000 shots), the FakeManila simulation (Fig. 11) recovers clear spectral structure: the two dominant peaks near $E \approx -5$ and $E \approx -2$ emerge above the noise floor, in stark contrast to the flat $n=20$ response (Fig. 8). This confirms that Trotter depth reduction combined with t_{rms} tuning is the key enabler for hardware-level execution.

4.8 Execution on Real IBM Quantum Hardware

Although the depth-reduction parameters were optimised using the 5-qubit Manila noise profile, the final hardware execution was mapped onto a 5-qubit connected subgraph of the 156-qubit `ibm.kingston` processor, accessed via the IBM Quantum Platform [4]. The circuit parameters are those validated on FakeManila: $N=5$, $n=3$, $t_{\text{rms}}=1.0$, 2000 shots, 100 scan points.

Figure 12 shows raw (red) and mitigated (blue) spectra. The raw output preserves partial structure around $E \in [-6, -2]$ but is substantially degraded. Two mitigation strategies are applied:

- **Dynamical decoupling (DD):** identity-equivalent pulse sequences during idle periods refocus low-frequency dephasing.

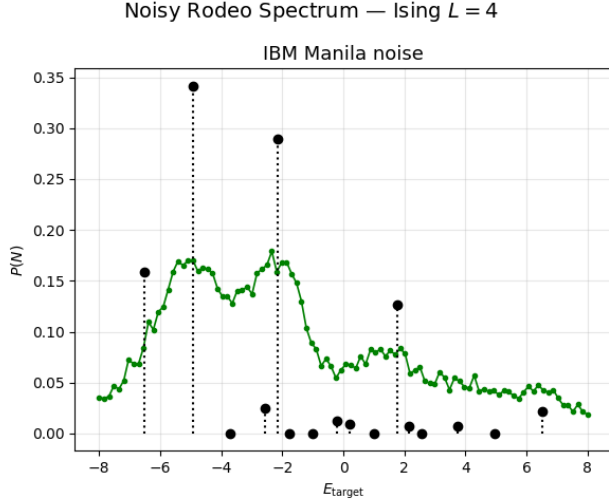


Figure 11: FakeManila simulation with reduced depth: $n=3$, $N=5$, $t_{\text{rms}}=1.0$, 2000 shots, 100 scan points. Spectral features are now clearly visible compared to the $n=20$ case (Fig. 8).

- **Pauli twirling:** random Pauli conjugation of two-qubit gates converts coherent errors into stochastic noise.

The mitigated spectrum shows improved maximum-peak probability and recovers the feature near $E \approx -1.8$ that was lost without mitigation; however, the peak near $E \approx -5$ is not recovered. With $G_{2q} \approx 300$ and $\sim 1\%$ per-gate error, the coherent survival is $(0.99)^{300} \approx 5\%$, leaving limited signal for mitigation to restore.

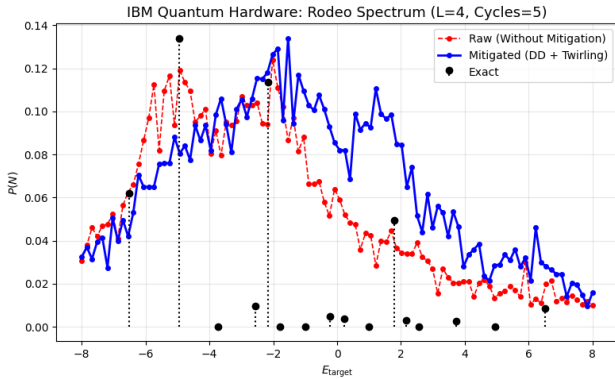


Figure 12: Rodeo spectrum from real IBM quantum hardware ($N=5$, $n=3$, $t_{\text{rms}}=1.0$, 2000 shots). Red: raw output without error mitigation. Blue: mitigated with dynamical decoupling (DD) and Pauli twirling. Black stems: exact eigenvalues. The dominant spectral structure changes with a slight improvement in the maximum peak, but the mitigated curve also loses information of other peaks present in the ideal spectrum.

4.9 Application to Other Hamiltonians

The Rodeo circuit is structurally independent of the target Hamiltonian: the same ancilla protocol (Fig. 1) applies to any H whose time evolution can be compiled into gates. The following subsections validate this generality on three additional models, using Qiskit’s `PauliEvolutionGate` for the controlled evolution.

4.10 Discrete Dirac Spectrum

Applying the Rodeo filter to the lattice Dirac Hamiltonian (Eq. 12, $L=4$, $m=1.0$) yields spectral peaks consistent with the dispersion relation (13). Both the positive-energy (matter) and negative-energy (antimatter) branches are resolved (Fig. 13). In this first-quantised formulation, both branches are intrinsic eigenvalues of the Hamiltonian matrix and are captured by scanning E_{tgt} across the full energy range—no second-quantisation formalism is required. The noiseless simulation places the peaks at the exact analytic positions for two distinct mass parameters.

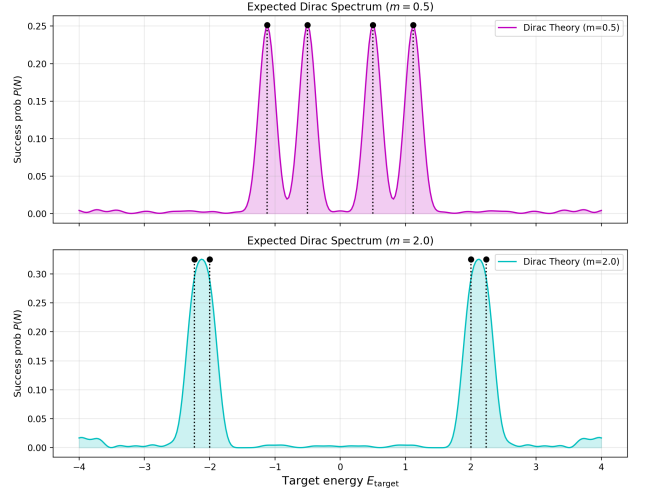


Figure 13: Expected Rodeo spectrum for the 1D discrete Dirac Hamiltonian under two distinct mass parameters ($m = 0.5$ top, $m = 2.0$ bottom), showing accurate anti-matter and matter eigenvalue extractions.

The Rodeo filter is next applied to the 1D isotropic Heisenberg XXX anti-ferromagnet:

$$H_{\text{XXX}} = J \sum_{i=1}^{L-1} (X_i X_{i+1} + Y_i Y_{i+1} + Z_i Z_{i+1}), \quad (19)$$

where $\vec{\sigma}_i \cdot \vec{\sigma}_{i+1}$ is the $\text{SU}(2)$ -invariant exchange interaction, generating multiplet degeneracies. For $L=3$ ($J=1.0$), the spectrum contains a quartet ground state ($S_{\text{tot}} = 3/2$) and two doublets ($S_{\text{tot}} = 1/2$), for four distinct energy levels. The filter resolves all four (Fig. 14), including the degenerate multiplet at $E = -3J$.

4.11 2D Harmonic Oscillator

To validate the n -dimensional factorisation (Eq. 5), we simulate a 2D harmonic oscillator on an $L=100$ per-dimension grid (10000×10000 matrix). The extracted

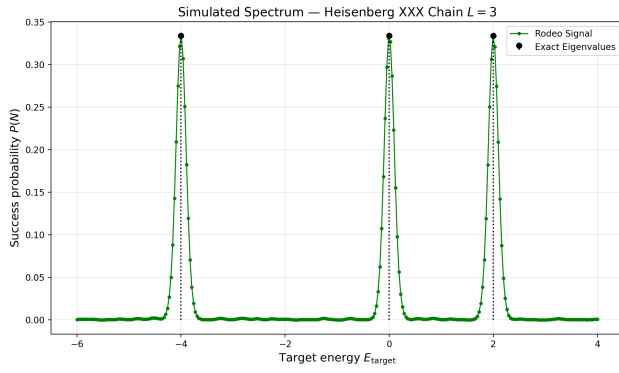


Figure 14: Rodeo spectrum for the $L = 3$ Heisenberg XXX model. Continuous red lines show the simulated filter, exactly intersecting theoretical eigenvalues (stems).

peaks align with the theoretical $(n + \frac{1}{2})\omega$ values with a ground-state relative error of 0.06% (Fig. 15).

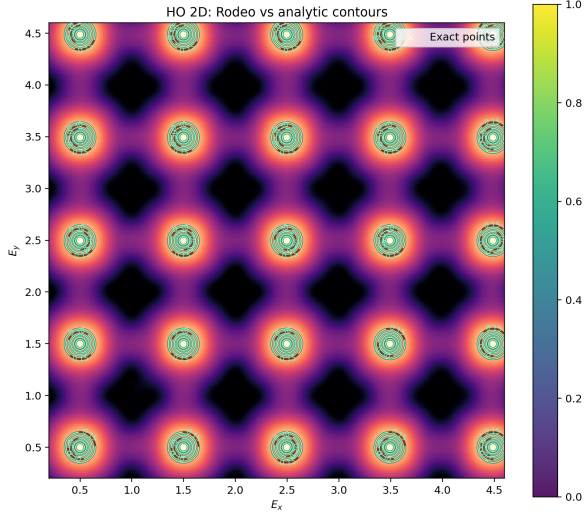


Figure 15: High-resolution 2D Rodeo spectrum for a Harmonic Oscillator ($L = 100$). The cyan crosses indicate exact matrix eigenvalues, while white circles show the theoretical continuum values $(n + 1/2)\omega$.

4.12 2D Dirac Hamiltonian

The 2D lattice Dirac Hamiltonian H_D^{2D} (Eq. 14, $L=12$ per dimension, $m=1.0$) is scanned in the (E_x, E_y) plane using the factorised filter (Eq. 5). The heatmap (Fig. 16) resolves the mass gap at $|E|=m$ and the symmetric matter/antimatter contours predicted by Eq. (15).

Dirac wave-packet dynamics. The Rodeo-identified eigenvalues are used to construct time-domain dynamics via spectral decomposition. Given eigenstates $H|\phi_n\rangle = E_n|\phi_n\rangle$ and initial state $|\psi(0)\rangle = \sum_n c_n |\phi_n\rangle$, the time evolution is

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle = \sum_n c_n e^{-iE_n t} |\phi_n\rangle. \quad (20)$$

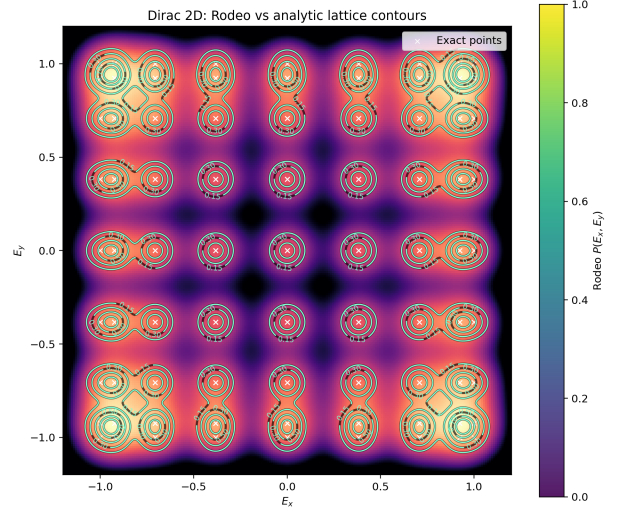


Figure 16: 2D Rodeo heatmap for the discrete Dirac model in the (E_x, E_y) plane ($L = 12$ per dimension, $m = 1.0$), with cyan analytic contours built from the exact discrete lattice momentum nodes.

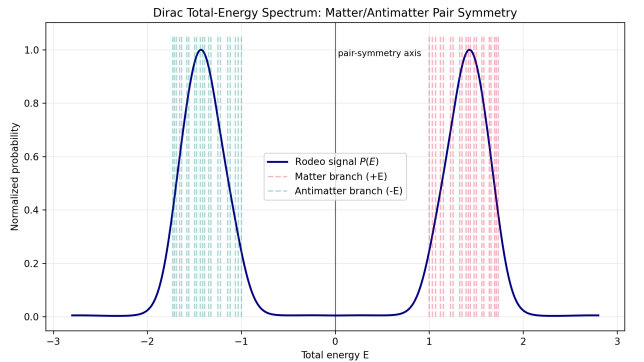


Figure 17: Dirac spectrum versus total energy E , showing symmetric positive (matter) and negative (antimatter) branches around $E = 0$, suitable for pair-creation analysis.

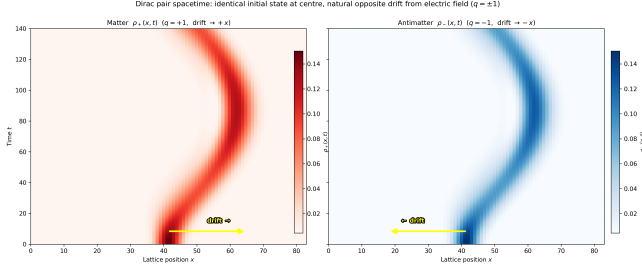


Figure 18: Spacetime density maps for a Dirac matter–antimatter pair under a uniform electric field, shown in *natural* (un-mirrored) coordinates. Both packets share the same initial Gaussian state centred at $x_0 = (L-1)/2$ with $k_0 = 0$; the only difference is the sign of the charge ($q = \pm 1$). The Rodeo-active positive- and negative-energy eigenstates are propagated with e^{-iHt} under the respective charged-sector Hamiltonians $H_{\pm}$. Yellow arrows indicate the direction of field-induced drift: matter drifts to $+x$, antimatter to $-x$, as dictated by $F = qE$.

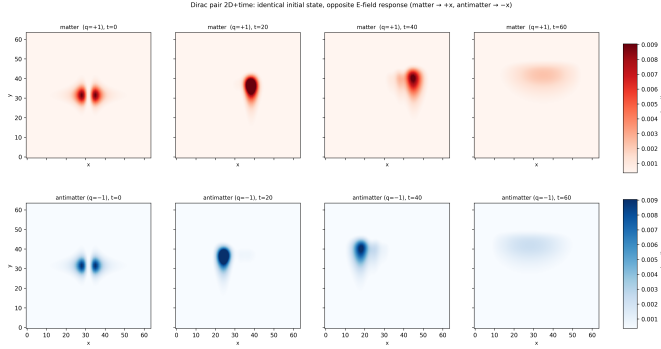


Figure 19: 2D+time Dirac pair dynamics under a uniform electric field. Both matter ($q = +1$) and antimatter ($q = -1$) start from *identical* conditions: centred Gaussian at (x_0, y_0) with $k_0 = 0$. The Rodeo-active modes are BZ-wrapped as $c_x = qEt$, producing naturally opposite group velocities and visible opposite drift without coordinate mirroring. Both rows share the same colorscale (global $\rho_{\max}$, floor 4%) for direct comparison.

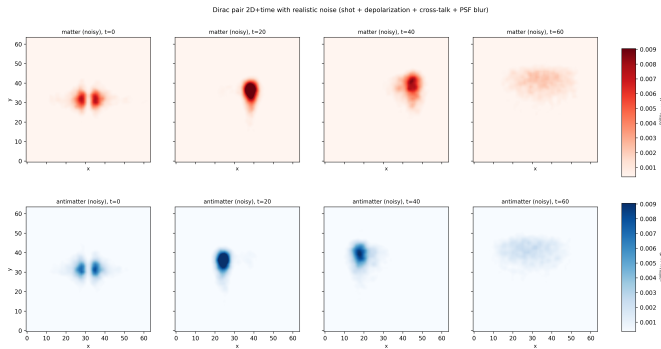


Figure 20: Standard noisy counterpart to Fig. 19: the Rodeo four-stage noise model (shot + depol + cross-talk + PSF) applied jointly to both channels. For a systematic five-level degradation study see Fig. 21.

Matter and antimatter components are obtained by spectral projection:

$$P_+ = \sum_{E_n > 0} |\phi_n\rangle \langle \phi_n|, \quad P_- = \sum_{E_n < 0} |\phi_n\rangle \langle \phi_n|, \quad (21)$$

$$|\psi_{\pm}(0)\rangle = P_{\pm} |\psi(0)\rangle, \quad \rho_{\pm}(x, t) = \sum_{s=\uparrow, \downarrow} |\langle x, s | \psi_{\pm}(t) \rangle|^2. \quad (22)$$

Both packets start from the same initial Gaussian centred at the lattice midpoint:

$$|\psi_s(0)\rangle = \mathcal{N} \sum_{x=0}^{L-1} \exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right] e^{ik_0 x} \times |x\rangle \otimes |s\rangle, \quad (23)$$

with $(x_0, \sigma, k_0) = ((L-1)/2, 3.6, 0)$, $|s\rangle = |\uparrow\rangle$ for matter and $|\downarrow\rangle$ for antimatter. The combined state $|\psi(0)\rangle = (|\psi_{\uparrow}(0)\rangle + |\psi_{\downarrow}(0)\rangle)/\sqrt{2}$ is projected by $P_{\pm}$ before propagation. The spectral coefficients are

$$c_n = \langle \phi_n | \psi(0) \rangle, \quad c_n^{(+)} = \langle \phi_n^{(+)} | \psi_+(0) \rangle, \quad c_n^{(-)} = \langle \phi_n^{(-)} | \psi_-(0) \rangle, \quad (24)$$

where $\{\phi_n^{(\pm)}\}$ denote the positive- and negative-energy eigenstates selected from the Rodeo-active spectral windows. The plotted centroid is the first moment

$$\bar{x}_{\pm}(t) = \frac{\sum_{x=0}^{L-1} x \rho_{\pm}(x, t)}{\sum_{x=0}^{L-1} \rho_{\pm}(x, t)}. \quad (25)$$

The connection to the Rodeo method is direct: the energy filter

$$F_N(E_{\text{tgt}}; E_n) = \prod_{k=1}^N \cos^2\left(\frac{(E_n - E_{\text{tgt}})t_k}{2}\right) \quad (26)$$

selects eigencomponents near E_{tgt} ; the packet dynamics in Figs. 18–22 are the time-domain continuation of this Rodeo-selected energy content.

To include an electric field, we add $V_E(x) = -qE(x - x_{\text{mid}})$ with $x_{\text{mid}} = (L-1)/2$ and evolve the matter ($q=+1$) and antimatter ($q=-1$) sectors under $H_{\pm} = H_D + V_E$.

Why is opposite drift expected? The force on a charge q in a uniform field $\mathcal{E}$ is $F = q\mathcal{E}$: a positron accelerates along the field, an electron against it [7]. On the lattice the same physics holds: matter and antimatter eigenstates carry opposite effective charges and acquire opposite group velocities under the same applied field. In the simulation both packets share identical initial conditions; only the sign of q in V_E differs, reversing the potential slope and producing the opposite drifts visible in Figs. 18–22—a direct signature of the Dirac charge-conjugation symmetry confirmed by the Rodeo-identified spectral branches.

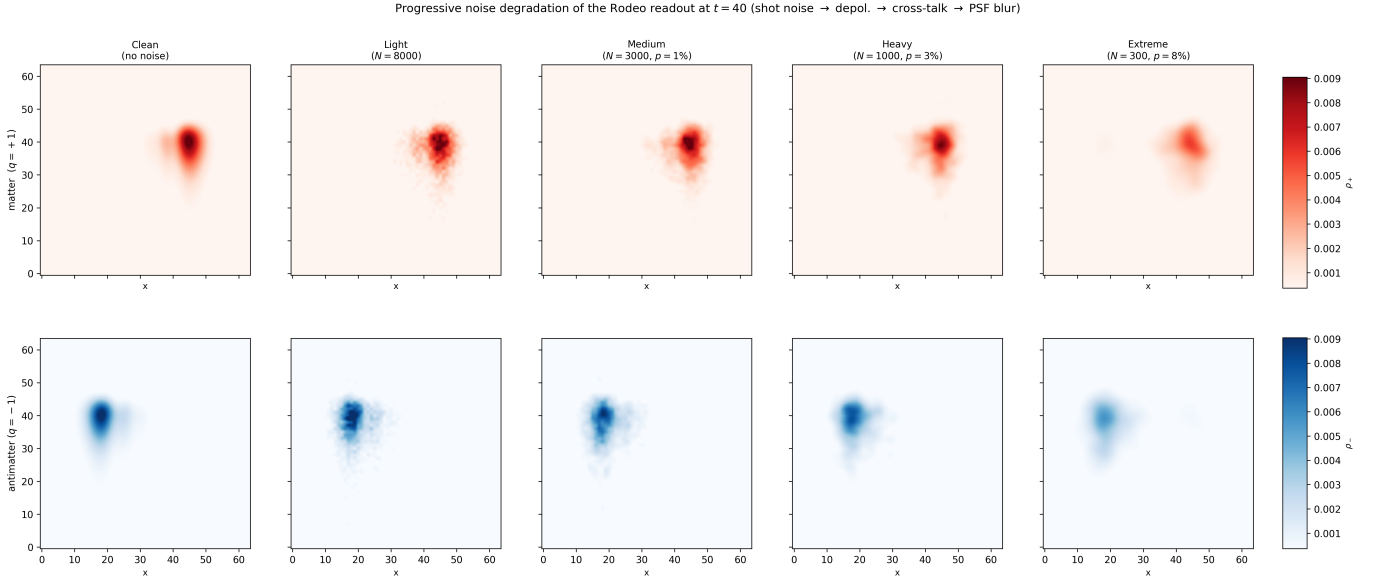


Figure 21: Progressive Rodeo readout degradation at $t = 40$ (where the matter/antimatter separation is clearly established). Five noise levels are shown left-to-right: clean $\rightarrow$ light shot noise ($N = 8000$) $\rightarrow$ medium ($N = 3000$, depol 1%) $\rightarrow$ heavy ($N = 1000$, depol 3%) $\rightarrow$ extreme ($N = 300$, depol 8%, cross-talk 8%, PSF blur $\sigma = 1.8$). The global colorscale is fixed to $\rho_{\max}$ of the clean panel; both rows (matter red, antimatter blue) share the same scale. The figure demonstrates that the Rodeo spatial signal retains the qualitative charge separation up to medium noise, and is fully washed out only in the extreme regime.

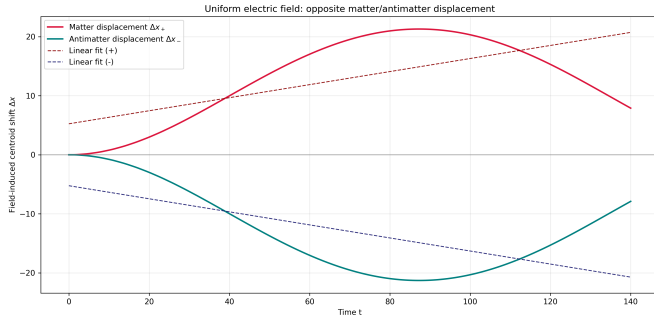


Figure 22: Field-induced centroid shifts under a uniform electric field. We plot $\Delta x_+(t) = x_+^{(E)}(t) - x_+^{(E=0)}(t)$ and the charge-weighted antimatter coordinate $\Delta x_-(t) = -(x_-^{(E)}(t) - x_-^{(E=0)}(t))$, which makes the opposite-sign electric response explicit. Dashed lines are linear fits, showing near-linear drift with opposite slopes.

4.13 2D Ising and 2D Heisenberg XXX

As a final validation, small 2×2 lattices (open boundaries) are simulated for both the TFIM and the Heisenberg XXX model. Figures 23 and 24 show that the Rodeo signal reconstructs the low-energy structure and aligns with exact diagonalisation. These results are obtained from full quantum-circuit Rodeo simulations in Qiskit Aer (ancilla superposition, controlled e^{-iHt} , phase kick, post-selection), not from the analytic filter approximation.

4.14 Ferromagnetic-to-Paramagnetic Transition

The Rodeo filter can also operate as a swept spectral probe: scanning a physical control parameter while measuring the energy axis reveals how the spectrum—and derived observables—evolve.

We apply this to the $L=8$ TFIM (OBC),

$$H = -J \sum_{i=1}^{L-1} Z_i Z_{i+1} - h \sum_{i=1}^L X_i, \quad (27)$$

sweeping h/J from the ferromagnetic ($h \ll J$) to the paramagnetic ($h \gg J$) regime. A small longitudinal bias $\delta_z = 0.015J$ breaks the $\mathbb{Z}_2$ symmetry so that the longitudinal magnetisation $\langle M_z \rangle = \langle \sum_i Z_i \rangle$ is well-defined.

Rodeo spectrogram (Fig. 25a). The Gaussian-approximation filter

$$P(E_{\text{tgt}}) = \frac{1}{d} \sum_{n=1}^d \exp \left[-\frac{N t_{\text{rms}}^2}{4} (E_n - E_{\text{tgt}})^2 \right], \quad (28)$$

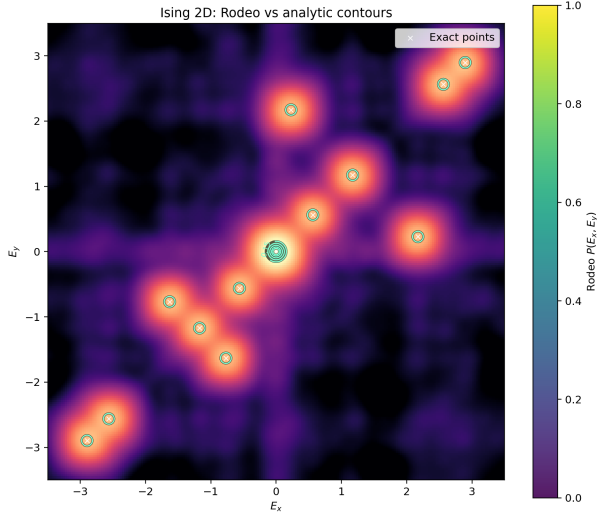


Figure 23: 2D Ising map in the (E_x, E_y) plane for a 2×2 lattice. Cyan markers indicate analytic eigen-branches projected onto (E_x, E_y) .

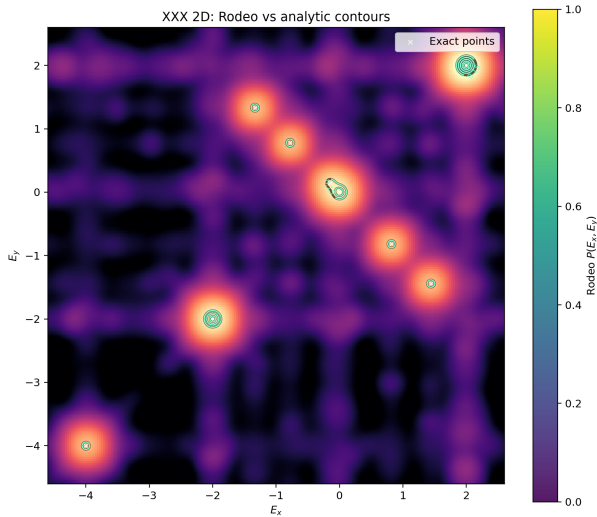


Figure 24: 2D Heisenberg XXX map in the (E_x, E_y) plane for a 2×2 lattice. Cyan markers indicate analytic eigen-branches projected onto (E_x, E_y) .

with $N=14$ and $t_{\text{rms}}=1.6$, is evaluated over a dense $(h/J, E_{\text{tgt}})$ grid. Bright ridges track the exact eigenvalues (cyan overlay); the yellow dashed curve marks the Rodeo-identified ground-state energy, which follows the exact E_0 closely throughout the sweep.

Ground-state magnetisation (Fig. 25b). Two curves are compared: exact diagonalisation (solid red) and Rodeo-identified ground state (dashed yellow), the latter selected as the eigenstate matching the lowest Rodeo peak. Both overlap, confirming correct ground-state identification. The order parameter $|\langle M_z \rangle|/L$ drops from ≈ 1 (ferromagnetic) to ≈ 0 (paramagnetic) near $h \approx J$, the quantum critical point.

Magnetic susceptibility (Fig. 25c). $\chi = |d\langle M_z \rangle/dh|$ peaks sharply at $h \approx J$, with exact and Rodeo-derived curves in excellent agreement. For a finite chain ($L=8$) the peak is rounded; it would diverge in the thermodynamic limit.

5 Conclusions

This work demonstrates a complete pipeline—from algorithm design through noise characterisation to real-hardware execution—for the Rodeo quantum-circuit eigenvalue filter.

In the noiseless regime the filter resolves the full $L=4$ Ising spectrum with $N=10$ cycles, converging at ~ 1000 shots per scan point (Figs. 2–5). However, the measured signal degrades exponentially with circuit depth under any realistic noise model: the coherent survival probability $(1-p)^{G_{2q}}$ drops below 10^{-9} at $G_{2q} \approx 2000$ (the $n=20$ operating point), and the IBM FakeManila simulation confirms a flat spectral response at this depth (Fig. 8).

The principal engineering result is the identification of a viable parameter window. Reducing Trotter steps to $n=3$ alone does not suffice—at $t_{\text{rms}}=5$ the filter cannot distinguish true peaks from Trotter-induced artefacts (Fig. 9). Jointly reducing t_{rms} to 1 recovers meaningful spectral structure in the FakeManila simulation (Fig. 11), establishing the parameter set ($n=3$, $N=5$, $t_{\text{rms}}=1$) as a feasible operating point.

On real hardware (`ibm_kingston`, 156 qubits), the dominant spectral feature near $E \approx -5$ is partially preserved in the raw output (Fig. 12). DD + Pauli twirling recovers the peak near $E \approx -1.8$ but loses the $E \approx -5$ feature, consistent with the $\sim 5\%$ coherent survival at $G_{2q} \approx 300$. Current device fidelities therefore constrain the method to low-depth, low-resolution configurations.

The algorithm’s model-independence is confirmed by applying the *same* Rodeo circuit to the discrete Dirac Hamiltonian (1D and 2D), the Heisenberg XXX chain, and the 2D harmonic oscillator. The Dirac time-domain continuation (Figs. 18–19) displays the expected opposite drift of matter and antimatter under an electric field, and a parameter sweep of the TFIM tracks the ferromagnetic-to-paramagnetic transition (Fig. 25).

The central lesson is quantitative: the depth–accuracy

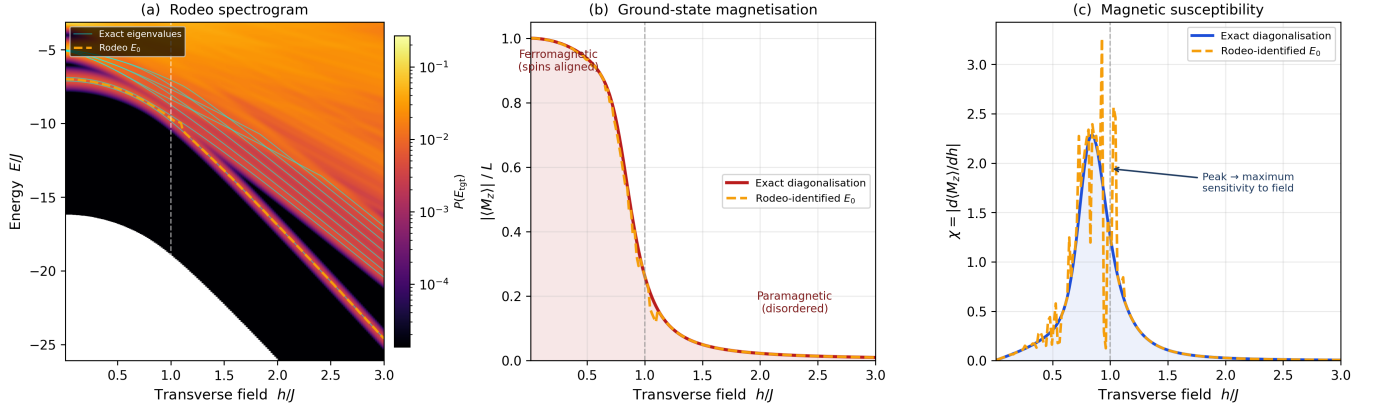


Figure 25: Ferromagnetic-to-paramagnetic transition in the TFIM ($L=8$, OBC): **Rodeo filter vs. exact diagonalisation**. (a) Rodeo spectrogram $P(h/J, E_{tgt})$: bright ridges trace eigenvalue branches; cyan curves are exact eigenvalues, yellow dashed curve is the Rodeo-identified ground-state energy E_0 . (b) Longitudinal magnetisation $|\langle M_z \rangle|/L$: solid red = exact, dashed yellow = Rodeo-identified ground state. Both overlap, confirming the Rodeo circuit correctly tracks the ferromagnetic-to-paramagnetic crossover. (c) Magnetic susceptibility $\chi = |d\langle M_z \rangle/dh|$ (same colour convention), peaking at $h \approx J$ —the quantum analogue of divergent susceptibility near the Curie temperature.

trade-off is the binding constraint for any Rodeo implementation on NISQ hardware. Meaningful spectra require $G_{2q} \lesssim 300\text{--}500$, which at current gate fidelities limits the method to small systems ($L \leq 5$), few Trotter steps ($n \leq 3$), and moderate cycle counts ($N \leq 5$). Overcoming this limit will require either substantially improved gate fidelities or advanced error-mitigation techniques (zero-noise extrapolation, probabilistic error cancellation), together with higher-order Trotter–Suzuki decompositions that reduce the gate count at fixed approximation accuracy.

References

- [1] K. Choi, D. Lee, J. Bonitati, Z. Qian, and J. Watkins, “Rodeo Algorithm for Quantum Computing,” *Phys. Rev. Lett.* **127**, 040505 (2021); arXiv:2009.04092.
- [2] Qiskit contributors, “Qiskit: An open-source framework for quantum computing,” 2024. <https://qiskit.org>.
- [3] H. F. Trotter, “On the product of semi-groups of operators,” *Proc. Amer. Math. Soc.* **10**, 545 (1959).
- [4] IBM Quantum, “IBM Quantum Platform,” 2024. <https://quantum.ibm.com>.
- [5] V. de O. Raimundo, “Quantum algorithm applied to the generalised eigenvalue problem (GEP),” B.Sc. thesis (Electronic Engineering), Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, Brazil, 2025. Advisor: Prof. Dr. Tobias Frederico.
- [6] C. Cohen-Tannoudji, B. Diu, and F. Laloë, *Quantum Mechanics*, Vols. 1–2, Wiley, 1977.

- [7] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Westview Press, 1995.